scientific reports

Check for updates

OPEN

Machine learning-based academic performance prediction with explainability for enhanced decision-making in educational institutions

Wesam Ahmed[1], Mudasir Ahmad Wani[2], Pawel Plawiak[3,4], Souham Meshoul[5]✉, Amena Mahmoud[6] & Mohamed Hammad[2,7]✉

Education is crucial for the growth of effective life skills and the allocation of needed resources. Higher education institutions are adopting advanced technologies, such as artificial intelligence (AI), to enhance traditional teaching methods. Predicting academic performance has become increasingly important, improving university rankings and expanding student opportunities. This study addresses challenges in performance analysis, quality education delivery, and student evaluation through machine learning (ML) models. Ten regression models including K-Nearest Neighbors Regressor, Linear Regression, CatBoost, XGBoost, AdaBoost, and ensemble voting regression (VR) algorithm based on the top five heterogeneous regressors as base models are employed to predict academic outcomes. Two datasets with distinct feature sets and sizes were used to evaluate the generalizability of the models. The first dataset comprises 10,000 samples and six features focused on study behaviors, prior performance, and extracurricular activities. The second dataset includes 6,607 records and 20 features encompassing academic habits, demographic attributes, and institutional factors such as attendance, teacher quality, and parental involvement. Best model performance goes to the linear regression in standalone ML models. Then, the proposed ensemble VR model was built using weighted averages based on the performances of the base models. The local interpretable model-agnostic explanations (LIME) and SHapley Additive exPlanations (SHAP) are then used to explain the predictions produced by the proposed ensemble VR model. For the first dataset, the VR model achieved an RMSE of 0.1050, MAE of 0.0837, and R^2 of 0.9890. On the second, more complex dataset, the VR model also performed best with an R^2 of 0.7716 using the full feature set, highlighting its robustness and adaptability across diverse academic contexts. These results offer actionable insights for educators, administrators, and policymakers to better understand student performance drivers and support data-informed educational strategies.

Keywords Academic performance prediction, Machine learning models, Voting regression, Educational decision-making, XAI, Artificial intelligence in education

The academic performance of students is a critical measure of the quality of educational institutions. It reflects the quality of education and influences an institution's reputation, ability to attract prospective students, and

[1]Department of Information Technology, Faculty of Computers and Artificial Intelligence, Hurghada University, Hurghada, Egypt. [2]EIAS Data Science Lab, College of Computer and Information Sciences, Prince Sultan University, Riyadh 11586, Saudi Arabia. [3]Department of Computer Science, Faculty of Computer Science and Telecommunications, Cracow University of Technology, Warszawska 24, Krakow 31-155, Poland. [4]Institute of Theoretical and Applied Informatics, Polish Academy of Sciences, Bałtycka 5, Gliwice 44-100, Poland. [5]Department of Information Technology, College of Computer and Information Sciences, Princess Nourah bint Abdulrahman University, P.O. Box 84428, Riyadh 11671, Saudi Arabia. [6]Computer Science Department, Faculty of Computers and Information, Kafrelsheikh University, Kafr El Sheikh, Egypt. [7]Department of Information Technology, Faculty of Computers and Information, Menoufia University, Shebin El Kom 32511, Egypt. ✉email: Sbmeshoul@pnu.edu.sa; mhammad@psu.edu.sa; mohammed.adel@ci.menofia.edu.eg

overall success. Higher education institutions aim to maintain high standards of teaching and learning to ensure their graduates meet industry expectations and societal needs. Certain institutions prioritize their reputation rather than the quality of education[1]. Academic performance is the most critical indicator of the quality of college students, and it is the primary criterion used by universities to assess students' academic abilities and monitor the quality of academic instruction[2]. Students' academic and practical performance are both evaluated during their academic years. Practical performance encompasses students' participation in on-campus scholarships, research competitions, and academic performance in various courses. Students admitted to the same college or university often have comparable test scores at the time of their enrollment. However, a progressive widening of performance gaps between students during their academic careers can be attributed to various factors[3]. Addressing these disparities is vital for improving overall student outcomes and fostering equitable educational opportunities.

The motivation for this work stems from the growing demand for data-driven methods in education to predict and enhance student performance. Machine learning (ML) has proven to be a valuable solution for this task, enabling the analysis of intricate patterns in educational data. Prior research has highlighted the effectiveness of various ML techniques, including decision trees, support vector machines (SVM), and boosting methods, in predicting academic outcomes[4]. Although ML algorithms have the potential to predict student academic performance, their efficacy and the impact of their implementation in educational environments are still unresolved[5]. There is a significant lack of understanding regarding the accuracy of these algorithms in predicting performance and the effective utilization of this data by educational institutions to make well-informed decisions. However, the previous works have limitations, including reliance on small datasets, lack of robust performance metrics, and inadequate comparison of multiple ML models within the same experimental framework. Additionally, some studies focus on isolated aspects of student performance, failing to provide a comprehensive analysis that accounts for various influential factors[6].

This study seeks to overcome these limitations by evaluating multiple ML regression models in a consistent experimental setting, using diverse and comprehensive performance metrics before implementing the voting model, enabling a comprehensive comparison of their respective performances. We have employed the voting algorithm to evaluate the risk of dropout and predict student performance. The predictions of multiple models are combined by voting regression algorithms, which capitalize on their strengths to generate a more precise final prediction. Voting regression can incorporate models trained using various algorithms, including Linear regression, Support Vector Regressor, and Ridge regression. This diversity can improve overall performance by capturing a broader spectrum of patterns and relationships in the data.

The main objective of this research study is to evaluate the efficacy and effects of the voting regression algorithm based on ML algorithms using all features and feature selection in educational environments and improve our understanding of model interpretability.

The key contributions of this study are highlighted below:

- A comprehensive comparative analysis of *nine* ML regression models is conducted using rigorous evaluation metrics to identify the best-performing model for predicting academic performance.
- We proposed a voting regression ensemble method that combines the predictions of the top *five* ML models to enhance the performance of student performance prediction.
- The study investigates the impact of feature selection on proposed model performance by comparing the results obtained with all features against a subset of features.
- The paper applies Local Interpretable Model-agnostic Explanations (LIME) and Shapley Additive Explanations (SHAP) to offer a deeper understanding of the voting model's predictions, which enhances the model's transparency and interpretability for educators and stakeholders.
- The findings offer actionable insights for educators and administrators, enabling data-driven approaches to enhance student success and institutional effectiveness.

Literature review

The continuous advancements in ML algorithms and the availability of student data have facilitated the development of more sophisticated academic performance prediction models. Predicting student success has become a vital area of research, as it enables higher education institutions to improve teaching strategies, optimize resource allocation, and identify at-risk students early. Several notable studies have explored various ML approaches for student performance prediction[7–13].

Haron et al.[7] introduced a hybrid methodology that integrates both quantitative and qualitative components to identify students at risk of unsatisfactory academic results. While effective in identifying high-risk students, the study's predictive capacity could be enhanced with the inclusion of more robust ML techniques. Vimarsha et al.[8] examined two modules: the first compares ML models for classifying teachers, while the second employs reinforcement learning (RL) for a multiclass student performance prediction model. Despite demonstrating RL's potential, the work lacks scalability and generalizability across diverse datasets. Ahmed[9] utilized a clustering-based approach with K-means and Davies-Bouldin metrics to analyze critical features influencing student performance. The study found that the SVM algorithm achieved the highest prediction accuracy but did not explore alternative ML models for comparison. Chen et al.[10] presented a bipartite network methodology combined with Louvain clustering for student performance forecasting. Although the algorithm demonstrated precise predictions, its complexity limits practical implementation. Liu et al.[11] developed a knowledge-tracing model incorporating an attention mechanism to predict success by leveraging student behavioral attributes and exercise characteristics. However, this approach does not fully consider interrelations among diverse knowledge areas, leading to suboptimal predictions. Mastour et al.[12] compared traditional and ensemble ML models for identifying at-risk and high-performing students. The study suggested incorporating university entrance examination data to enhance accuracy, which was not included in their analysis. Recent studies have emphasized the effectiveness of ensemble learning techniques in predicting student academic performance. For instance, Adewale et al.[2] presented a multilayered framework that integrates multiple machine learning models to enhance prediction accuracy in open and distance learning environments, demonstrating the potential of such approaches in educational settings. Xu and Kim[13] presented a mixed predictive technique employing the ant colony algorithm to determine the weight of different ML models. While innovative, the model's prediction accuracy requires improvement through better feature selection and optimization techniques.

Despite significant progress in student performance prediction, prior studies exhibit notable limitations. Many relied on small or narrowly scoped datasets, which limited the generalizability of their models. Additionally, some studies focused on isolated features, failing to account for the broader range of factors influencing academic performance. The lack of consistent evaluation metrics further hindered the ability to compare results across studies, and few comprehensive comparisons of multiple machine learning models under consistent experimental conditions.

Our study addresses these limitations through a rigorous and systematic approach. The proposed voting regression model enhances prediction accuracy and robustness by integrating the strengths of multiple models. Ensemble methods, such as voting regression, are less susceptible to overfitting than individual models, thereby increasing the reliability and generalizability of predictions to previously unseen data. The integration of LIME and SHAP for prediction interpretation can assist in addressing the "black box" nature of intricate machine learning models to improve model interpretability. Table 1 provides a detailed comparison of prior works on student performance prediction, highlighting their limitations and how our method addresses these shortcomings effectively.

Proposed methodology

This section outlines the methodological framework used to predict student performance. The process began with data preprocessing to prepare the dataset for analysis. Next, *nine* individual ML regression models were selected. Afterward, the performance of both the voting regressor and the base regressors was evaluated and compared using *three* different metrics on the validation dataset. Finally, the section presents an explanation and interpretation of the model's predictions. Figure 1 illustrates the overall architecture of the proposed student performance prediction system.

Data collection and preprocessing

There are *two* datasets utilized in this study. The first dataset provides a comprehensive representation of students' academic performance by including various features such as grades, Sleep Hours, and extracurricular activities. It comprises 10,000 rows and six features[14], including 'Hours Studied', 'Previous Scores', 'Extracurricular Activities', 'Sleep Hours', 'Sample Question Papers Practiced', and 'Performance Index'—which serves as the target variable reflecting overall academic achievement. The dataset, originally obtained from Kaggle, has been validated to contain no missing or duplicate records, ensuring its suitability for analysis. These features are a mix of numerical and categorical variables. 'Hours Studied', 'Previous Scores', 'Sleep Hours', and 'Sample Question Papers Practiced' are continuous numerical variables, while 'Extracurricular Activities' is a binary categorical feature (Yes/No). As shown in Table 2, each feature is described in detail to highlight its relevance. Preprocessing is a critical step for preparing the dataset for ML models. First, categorical features such as extracurricular activities are encoded using a label encoder to convert them into numerical values. Subsequently, numerical features are normalized using a standard scaler to normalize the data distribution and enhance the models' convergence during training. Table 3 illustrates some examples of the first dataset. Table 4 highlights the statistical analysis of the first dataset.

Study	Methodology	Advantages	Limitations	How Our Method Overcomes Limitations
Haron et al.[7]	Hybrid (quantitative + qualitative components)	Early risk identification	Relied on narrow datasets and lacked robust model evaluation metrics.	Used a comprehensive dataset and proposed voting model, with diverse features and standardized evaluation metrics (RMSE, MAE, R^2) for a holistic analysis.
Vimarsha et al.[8]	RL-based multiclass prediction model	Demonstrates RL's potential	Limited comparison scope and absence of diverse performance metrics.	Conducted a comparative analysis of nine regression models and proposed a voting model, with rigorous metrics to identify the most effective algorithm.
Ahmed[9]	K-means clustering with SVM	Identifies critical features; high accuracy with SVM	Focused on isolated features and lacked a comprehensive analysis of student performance trends.	Integrated diverse features influencing academic outcomes for a broader perspective and evaluated performance across multiple dimensions.
Chen et al[10].	Bipartite network with Louvain clustering	Precise prediction	Relied on a specialized clustering method without broader comparisons to other ML models.	Compared nine advanced regression models and proposed voting model. within a consistent experimental framework for robust performance evaluation.
Liu et al.[11]	Attention-based knowledge-tracing model	Leverages student behaviors and exercise characteristics	Did not consider interrelationships among knowledge factors or diverse predictive metrics.	Addressed interrelationships through feature diversity and employed standardized evaluation to assess model effectiveness comprehensively.
Mastour et al.[12]	Traditional vs. ensemble ML models	Ensemble models improve performance	Did not incorporate external factors such as entrance examination scores, limiting model accuracy.	Included diverse features and holistic factors to enhance predictive accuracy and provide actionable insights.
Xu and Kim[13]	Ant colony algorithm for model integration	Innovative approach to weight determination	The model's efficacy was limited by inconsistent evaluation and the absence of robust testing frameworks.	Established a consistent experimental framework with robust testing and comparative analysis across multiple regression models and proposed voting model.

Table 1. Summary of previous studies on student performance prediction.

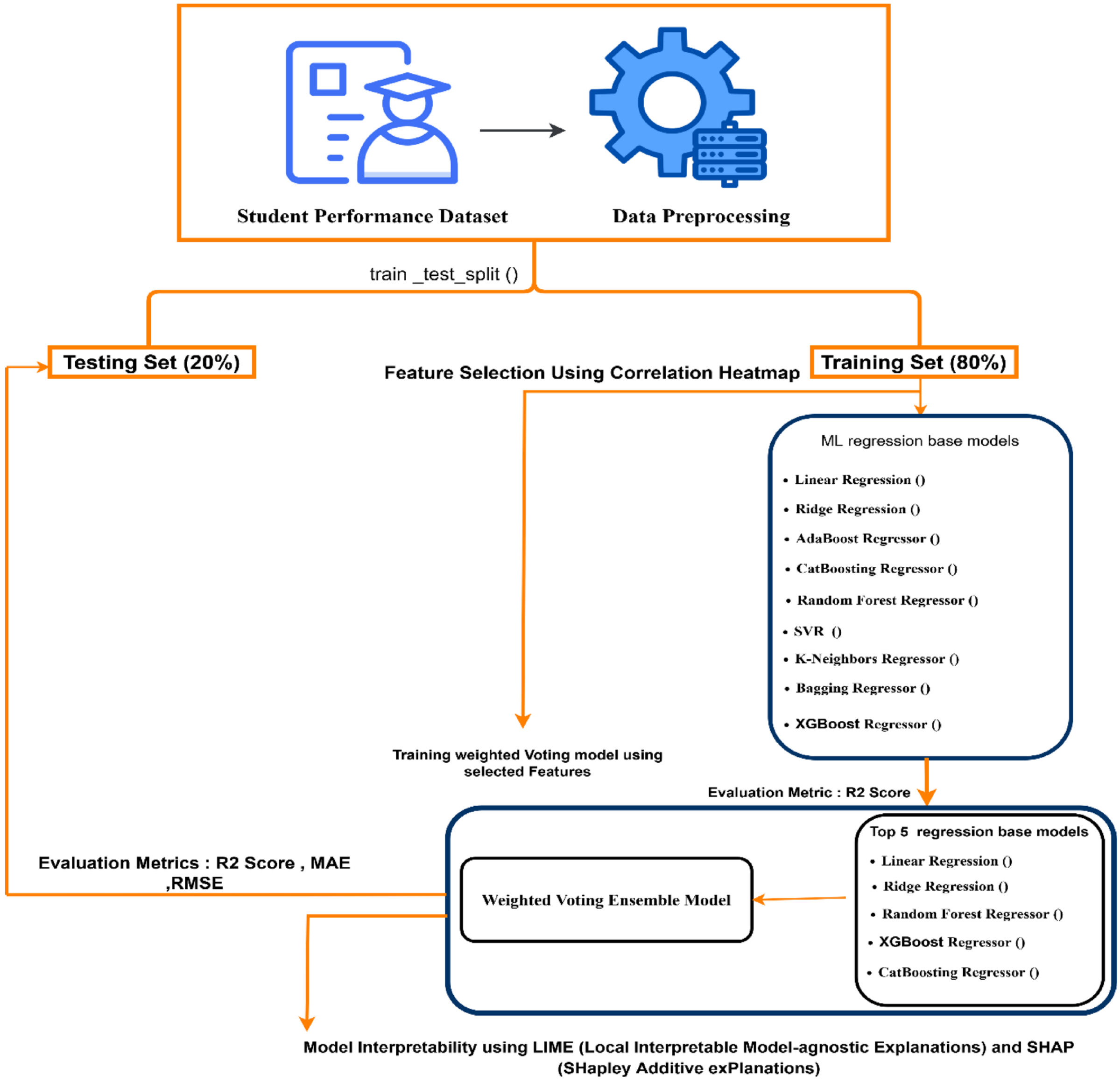

Fig. 1. Overall architecture of the proposed student performance prediction system.

Feature	Description
Extracurricular Activities	Indicates whether the student is involved in extracurricular activities (Yes/No)
Previous Scores	The average scores that students have achieved on their previous exams
Hours Studied	Daily average hours spent studying
Sleep Hours	Duration of average daily sleep
Performance Index	A numeric metric that has been calculated and summarizes the student's academic performance.

Table 2. Description of the features of the first dataset.

The second dataset is that student performance factors thoroughly examine the numerous variables that influence student exam performance[15]. It encompasses data regarding attendance, study practices, parental involvement, and other factors contributing to academic success. This dataset has 6,607 records and 20 features as shown in Table 5. There are seven numeric columns, including exam score, Attendance, and hours studied. There are 13 categorical columns, including gender, parental Involvement, access to resources, extracurricular activities, and teacher quality. The teacher quality, parental education level, and distance from home columns of the dataset contain missing values. The most frequently occurring value will be used to replace the missing values in each categorical column and with the mean value in numerical features. We identified 101 as an outlier

	Hours Studied	Previous Scores	Extracurricular Activities	Sleep Hours	Sample Question Papers Practiced	Performance Index
0	7	99	Yes	9	1	91.0
1	4	82	No	4	2	65.0
2	8	51	Yes	7	2	45.0
3	5	52	Yes	5	2	36.0
4	7	75	No	8	5	66.0

Table 3. Examples of the first dataset.

Parameters	count	mean	std	min	max	25%	50%	75%
Hours Studied	10000.0	4.9929	2.589309	1.0	9.0	3.0	5.0	7.0
Previous Scores		69.4457	17.343152	40.0	99.0	54.0	69.0	85.0
Sleep Hours		6.5306	1.695863	4.0	9.0	5.0	7.0	8.0
Sample Question Papers Practiced		4.5833	2.867348	0.0	9.0	2.0	5.0	7.0
Performance Index		55.2248	19.212558	10.0	100.0	40.0	55.0	71.0

Table 4. The statistical analysis of the first dataset.

Feature	Description
Hours_Studied	The amount of time dedicated to studying each week.
Motivation_Level	The degree of motivation of the student (Low, Medium, High)
Teacher_Quality	Level of teacher quality (Low, Medium, High)
Gender	Student's gender (Male or Female)
Exam_Score	Score of the final examination

Table 5. Description of various features of the second dataset.

	Hours Studied	Attendance	Parental Involvement	Access to Resources	Extracurricular Activities	Gender	Exam Score
0	23	84	Low	High	No	Male	67
1	19	64	Low	Medium	No	Female	61
2	24	98	Medium	Medium	Yes	Male	74
3	29	89	Low	Medium	Yes	Male	71
4	19	92	Medium	Medium	Yes	Female	70

Table 6. Examples of the second dataset.

value in the exam score column; therefore, we have replaced it with 100. All the categorical features are converted into numerical values. Table 6 illustrates some examples of the second dataset.

Machine learning algorithms

To predict the student academic performance, we implemented *nine* ML algorithms, including Linear Regression, K-Nearest Neighbors (KNN), XGBoost, CatBoost, AdaBoost, Random Forest, Support Vector Regression (SVR), Ridge Regression, and Bagging Regressor. These models were selected based on their effectiveness in regression tasks and their ability to capture diverse patterns in educational data. Below, we detail each algorithm's role in predicting student outcomes, along with their mathematical formulations.

1) Multi-Linear Regression (MLR).

MLR models the relationship between a dependent variable (final grade) and multiple independent features (e.g., attendance, past grades, extracurricular activities) using a linear approach[16]. The model minimizes the residual sum of squares (RSS) to optimize coefficients. Mathematical formulation is:

$$\widehat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_n x_n + \epsilon \tag{1}$$

.

where $\widehat{y}$ is the predicted grade, β_0 is the interception, β_i are coefficients, x_i are input features, and ϵ is the error term. In our study, MLR identifies the influence of each feature (e.g., attendance, grades, extracurricular participation) on the final grade. This algorithm serves as a baseline for comparing other regression methods.

2) K-Nearest Neighbors Regression (KNN).

KNN regression is an instance-based algorithm that predicts the output for a given input by averaging the target values of the K-nearest data points in the feature space[17,18]. The following is determined using a distance metric, such as Euclidean distance. In our method, KNN predicts the performance of a student from the average of the grade of KK, using the Euclidean distance for proximity, the most similar students in the feature space are from the average of the same students. KNN can formally represent:

$$\widehat{y}(x) = \frac{1}{K}\sum{}_{i=1}^{K} y_i \tag{2}$$

.

where y_i are the target values of the K nearest neighbors. KNN effectively captures localized trends, making it useful for identifying peer-influenced performance patterns.

3) XGBoost Regression.

XGBoost, or Extreme Gradient Boosting, is an advanced implementation of promoting shields that involves regularization to prevent overfitting and increase computational efficiency[19,20]. It constructs decision trees iteratively, optimizing an objective function at each step. XGBoost is applied in our model to exploit its ability to capture complex, non-linear relationships among features, thereby improving prediction accuracy for diverse student data. Mathematically it can represent as:

$$\mathcal{L}(\theta) = \sum{}_{i=1}^{n} l(y_i, \widehat{y_i}) + \sum{}_{k=1}^{K} \Omega(f_k) \tag{3}$$

.

where l is the loss function, Ω penalizes model complexity, and f_k represents trees. The grid search used to optimize the hyperparameters, and the optimal values, which are n_estimators = 100, max_depth = 3, learning_rate = 0.1, min_child_weight = 1, gamma = 0, subsample = 0.8.

4) CatBoost Regression.

CatBoost, short for "Categorical Boosting," is a gradient boosting algorithm tailored for categorical data[21]. CatBoost handles categorical features (e.g., course type, extracurriculars) natively, reducing preprocessing needs and overfitting. In our case, CatBoost enhances predictions by leveraging its capability to process categorical attributes, such as extracurricular activities, directly within the model. It can be represented mathematically as:

$$\widehat{y_i} = \sum{}_{j=1}^{M} \eta . h_j(x_i) \tag{4}$$

.

where h_j are the decision trees, and η is the learning rate.

5) AdaBoost Regression.

AdaBoost, or Adaptive Boosting, is an ensemble method that consecutively combines uncertain learners, such as decision trees, to enhance overall prediction accuracy[22]. Each subsequent learner focuses on correcting errors made by previous learners. For our model, AdaBoost captures intricate patterns in the data and provides an ensemble-based perspective on student performance trends. It can be represented as:

$$\widehat{y}(x) = \sum{}_{t=1}^{T} \alpha_t h_t(x) \tag{5}$$

.

where α_t are weights, and h_t are weak learners.

6) Random Forest Regressor.

An ensemble of decision trees that reduces variance by averaging predictions from multiple trees[23]. Random Forest mitigates overfitting and handles high-dimensional educational data well. It represents as:

$$\widehat{y}(x) = \frac{1}{B}\sum{}_{b=1}^{B} T_b(x) \tag{6}$$

.

where B is the number of trees, and $T_b(x)$ are individual tree predictions.

7) SVR.

SVR maps feature high-dimensional space to fit a hyperplane with maximum margin tolerance[24]. SVR is effective for small datasets with non-linear relationships. The mathematical formulation of it is as follows:

$$min\frac{1}{2}\parallel w \parallel^2 + C\sum {}_{i=1}^{n}\left(\xi_{\ i} + \xi_{\ i}^{\ *}\right) \text{ subject to } \mid y_i - \langle w, x_i\rangle - b \mid \leq \ \epsilon \ + \xi_{\ i} \tag{7}$$

.

8) Ridge Regression.

A regularized linear regression variant that prevents overfitting via L2 penalty[25]. Ridge improves stability in multicollinear features (e.g., correlated course grades). Mathematically represented as:

$$\widehat{y} = \underset{w}{\text{argmin}}\parallel y - Xw \parallel^2 + \alpha \parallel w \parallel^2 \tag{8}$$

.

Ridge improves stability in multicollinear features (e.g., correlated course grades).

9) Bagging Regressor.

Aggregates predictions from multiple base models (e.g., decision trees) trained on random subsets of data[26]. Bagging reduces variance and enhances generalization. It represented mathematically as:

$$\widehat{y}\left(x\right) = \frac{1}{M}\sum {}_{m=1}^{M}F_m\left(x\right) \tag{9}$$

.

10) Ensemble Voting Regressor (Proposed Model).

The proposed Ensemble Voting Regressor (VR) was designed to enhance prediction accuracy by strategically combining the strengths of multiple machine learning models while mitigating their individual limitations. Prior to model training, feature selection was conducted to eliminate redundant or non-predictive variables, ensuring computational efficiency without compromising performance. This process relied on correlation analysis, as visualized in Fig. 2, which depicts Pearson correlation coefficients between input features and the target variable (Performance Index) of the first dataset. The analysis revealed that Previous Scores exhibited an exceptionally strong positive correlation ($r = 0.92$), establishing it as the most influential predictor of academic performance. Hours Studied showed a moderate correlation ($r = 0.37$), while other features—such as Extracurricular Activities ($r = 0.02$), Sleep Hours ($r = 0.05$), and Sample Question Papers Practiced ($r = 0.04$)—demonstrated negligible relationships with the target. Consequently, only Previous Scores and Hours Studied were retained for the feature-selected subset, streamlining the model by removing noise from weakly correlated variables. Figure 3 depicts Pearson correlation coefficients between input features and the target variable (Exam Score) of the second dataset. The analysis revealed that Attendence exhibited an exceptionally strong positive correlation ($r = 0.58$), establishing it as the most influential predictor of exam score. Hours_Studied showed a moderate correlation ($r = 0.45$).

The VR model operates by aggregating predictions from the top *five* standalone algorithms, weighted according to their individual performance. For the feature-selected scenario, the ensemble incorporated Linear Regression, Ridge Regression, CatBoost, XGBoost, and Random Forest, with weights assigned as[1,7] to prioritize linear models, which outperformed others in preliminary tests. In the full-feature scenario, SVR replaced Random Forest due to its better handling of high-dimensional data, though the weighting scheme remained consistent. The weights were optimized via grid search to minimize the root mean squared error (RMSE), ensuring the ensemble's predictions balanced bias and variance effectively. Mathematically, the VR's prediction $\widehat{y}VR$ is derived as a weighted sum of base model outputs:

$$\widehat{y}VR = \sum {}_{i=1}^{5}w_i.\widehat{y_i} \text{ where } \sum w_i = 1 \tag{10}$$

.

where w_i represents the normalized weight of the *i*-th model. Figure 4 illustrates the three-stage architecture of our proposed Ensemble Voting Regressor, which strategically combines predictions from *five* carefully selected base models to achieve optimal performance in student academic prediction. In our implementation:

1) Transition Set (Base Models):

We employed *five* distinct regression models that demonstrated complementary strengths in preliminary testing:

- Linear Regression and Ridge Regression (prioritized for their interpretability and stable performance on linear relationships).
- CatBoost and XGBoost (selected for their ability to capture complex non-linear patterns).

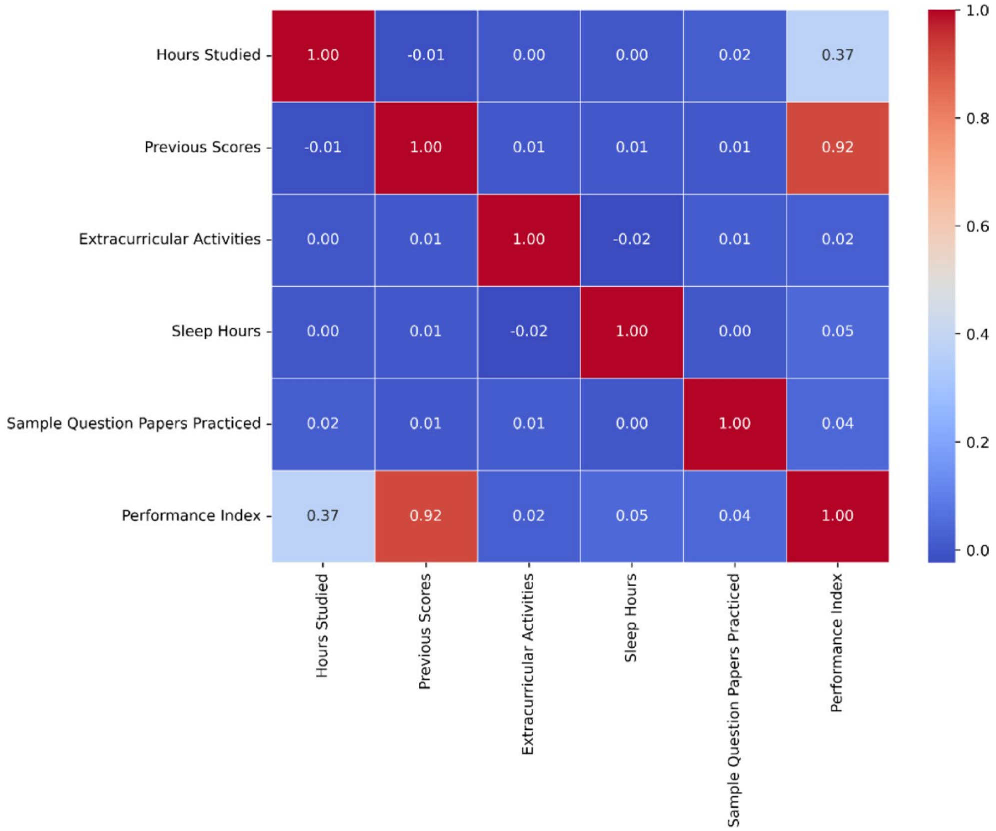

Fig. 2. Pearson correlation heatmap showing the relationship between input features and the target (Performance Index). Features with low correlation values (e.g., Extracurricular Activities) were excluded during feature selection.

- Random Forest (including for its robustness against overfitting).

Each model processes the input features (notably Previous Scores and Hours Studied after feature selection) to generate independent predictions (P_1 to P_5). The diversity of these models ensures that different aspects of student performance patterns are captured, from simple linear correlations to intricate non-linear relationships.

2) Voting Mechanism:

Our weighted voting scheme assigns importance values of[1,7] to the respective models, reflecting their relative predictive performance. This weighting:

- Strongly favors the linear models (7× weight each) due to their superior baseline performance.
- Moderately incorporates the tree-based models (1× weight each) to add nuanced pattern recognition.
- It was optimized through grid search to balance the strengths of each approac.

3) Final Prediction:

The ensemble's output combines these weighted predictions into a single, more accurate estimate of student performance. The architecture's effectiveness is particularly evident in how it handles edge cases - where one model might struggle, others compensate. For instance, when predicting performance for students with unusual study patterns (very high Hours Studied but average Previous Scores), the linear models provide stability while the tree-based models adjust for non-linear effects.

To address the "black-box" nature of ensemble methods, LIME and SHAP were employed[27]. LIME provided localized insights by approximating the VR model's behavior for individual predictions (e.g., identifying that a student's low Hours Studied value disproportionately reduced their predicted grade). SHAP offered a global perspective, quantifying the average contribution of each feature across the dataset. For instance, SHAP values confirmed that Previous Scores accounted for 62% of the VR model's output, aligning with the correlation heatmap. These interpretability tools not only validated the model's logic but also made its predictions actionable for educators—highlighting which factors (e.g., prior academic performance) were most critical for intervention.

The VR model achieved superior performance compared to all standalone algorithms, with an RMSE of 0.1050, MAE of 0.0837, and R^2 of 0.9890. Figure 5 further demonstrates the ensemble's stability, showing how the voting mechanism reduces error dispersion compared to base models like SVR and Random Forest, which

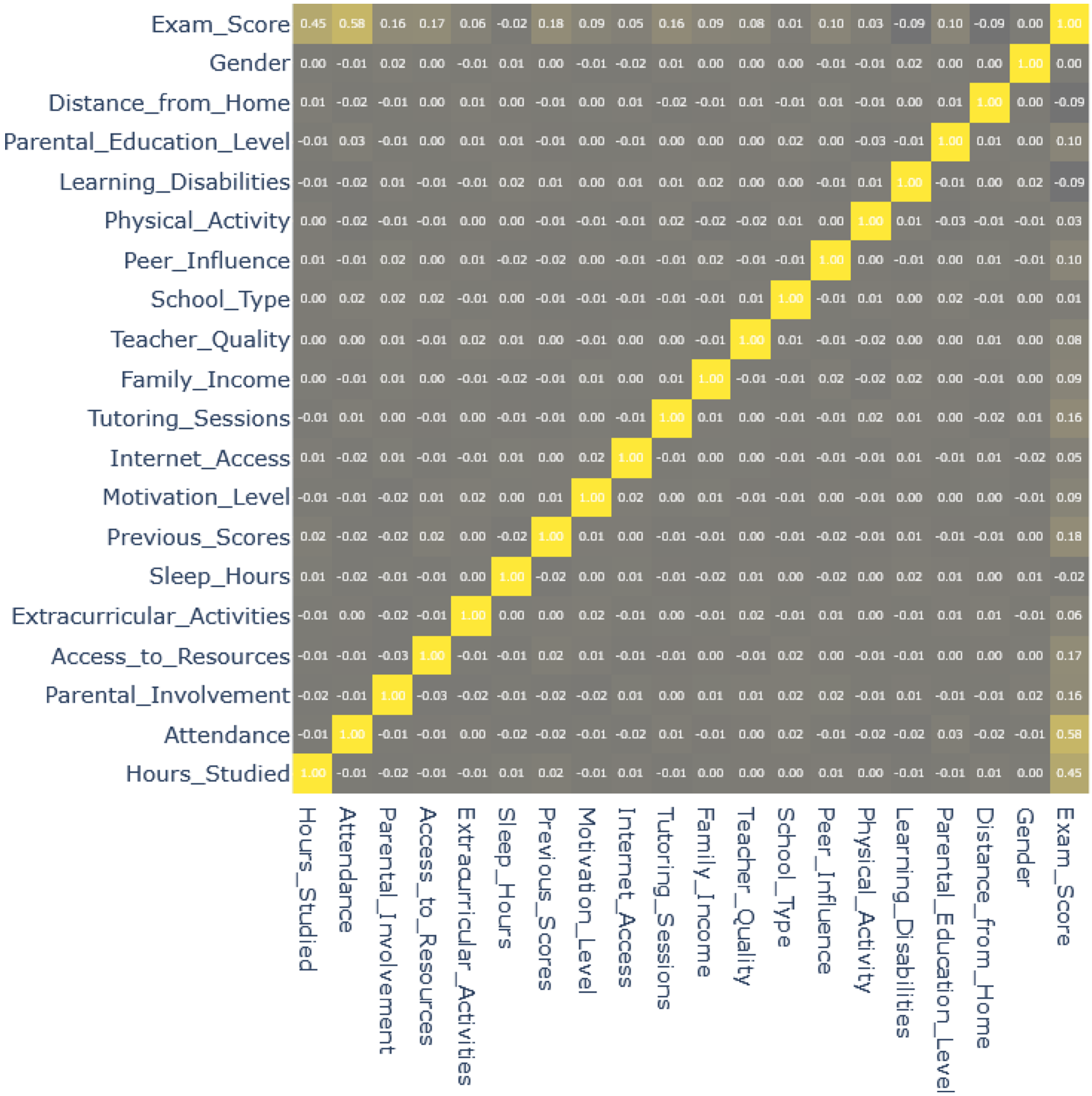

Fig. 3. Pearson correlation heatmap showing the relationship between input features and the target (Exam Score).

exhibited higher variance in predictions. This stability, coupled with the model's transparency through LIME/SHAP, positions the VR as a reliable tool for academic advisors seeking to identify at-risk students and tailor support strategies.

Table 7 highlights the trade-offs between interpretability, data handling, and predictive power, aiding educators in selecting suitable models for academic performance analysis.

Performance metrics

To evaluate the effectiveness of the proposed models, we use three standardized regression evaluation metrics[28,29]:

- Mean Absolute Error (MAE):

Calculates the average magnitude of prediction errors, expressed as:

$$MAE = \frac{1}{n}\sum{}_{i=1}^{n}|x_i - y_i| \tag{11}$$

MAE is particularly useful for interpreting the average prediction error in the same units as the dependent variable.

- Root Mean Square Error (RMSE):

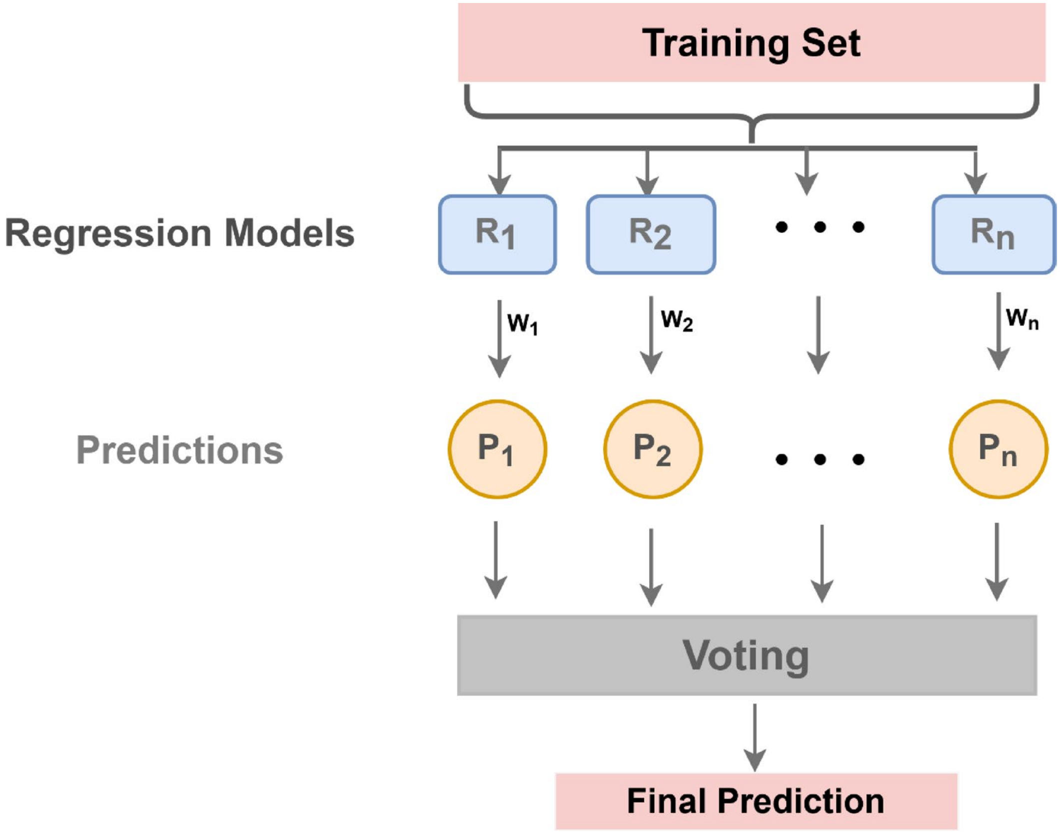

Fig. 4. Architecture of the proposed Ensemble Voting Regressor.

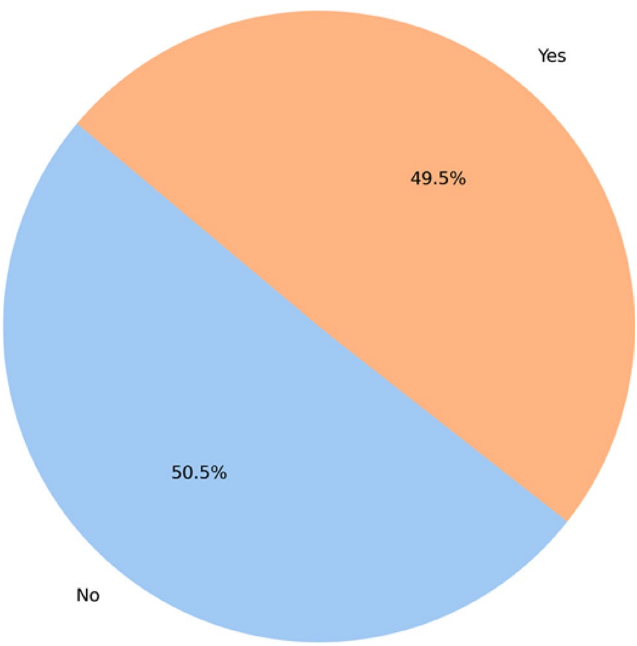

Fig. 5. Distribution of student participation in extracurricular activities.

Algorithm	Key Strength	Handling Categorical Data	Interpretability
MLR	Baseline linear modeling	Requires encoding	High
KNN	Local pattern detection	Requires encoding	Medium
XGBoost	Non-linear relationships + regularization	Requires encoding	Medium
CatBoost	Native categorical support	Direct handling	Medium
AdaBoost	Error-correcting ensembles	Requires encoding	Low
Random Forest	Robust to overfitting	Requires encoding	Medium
SVR	Small dataset efficiency	Requires encoding	Low
Ridge	Multicollinearity resistance	Requires encoding	High
Bagging	Variance reduction	Requires encoding	Medium
Ensemble VR	Optimal weighted predictions	Depending on base models	Low (LIME/SHAP used)

Table 7. Comparative summary of ML Algorithms.

Quantifies the standard deviation of prediction errors and penalizes larger errors more than MAE, calculated as:

$$RMSE = \sqrt{\frac{1}{n} \sum {}_{i=1}^{n} (x_{i-}\, y_i)2} \tag{12}$$

RMSE provides a sensitive measure for identifying significant deviations in predictions.

- R-Squared (R^2):

Indicates the extent to which the independent variables account for the variation observed in the dependent variable, and is determined using the formula:

$$R^2 = 1 - \frac{SSR}{SST} \tag{13}$$

where SSR represents the sum of squared residuals, and SST denotes the total sum of squares. Higher R^2 values indicate better model fit. By integrating these metrics, we ensure a comprehensive evaluation of model performance across various dimensions, facilitating reliable predictions and actionable insights.

Results and analysis

This section assesses the effectiveness of the developed regression models in forecasting student performance, both with selected features and using all available features. To ensure an unbiased and consistent evaluation, the datasets were divided into 80% for training and 20% for testing. The accuracy of the models' predictions was measured using *three* commonly used evaluation metrics such as MAE, RMSE, and R^2 Score. The results highlight the effectiveness of the proposed methodology and its potential for real-world applications. Figures 5 and 6, and 7 offer a deeper understanding of the dataset characteristics. Figure 5 shows a significant proportion of students who do not participate in extracurricular activities, which may impact overall performance due to reduced engagement in holistic development. Figure 6 illustrates that there is a positive correlation between the number of sleep hours and the performance index, indicating the importance of sufficient rest for academic success. Figure 7 shows the average performance index peaks when students practice nine question papers, suggesting that this is the optimal preparation level for achieving high performance.

We compared the performance of the base models when all features were utilized and when a subset of features was selected through feature selection. Table 8 summarizes the performance of the ten regression models using all features. Table 9 summarizes the performance of the proposed model using all features and cross validation using the first dataset. Figures 8 and 9 show the comparison between the models using all features. Figure 10 shows the actual and predicted values for the voting model using all features. Table 10 summarizes the performance of the ten regression models using feature selection on the first dataset. Table 11 summarizes the performance of the ten regression models using all features on the second dataset. Table 12 presents the performance of the same models after applying feature selection on the second dataset. Figures 11 and 12 show the comparison between the models using feature selection. Figure 13 shows the actual and predicted values for the voting model using feature selection.

The features' positive and negative influences on the decision-making process and outcome prediction were evaluated using SHAP in the proposed voting model. The significance of the SHAP feature for the trained proposed voting model is illustrated in Fig. 14, which illustrates the average impact on model output, where the previous scores are the most influential feature, approximately 0.67. Figure 15 presents a summary plot displaying the calculated SHAP values for each instance in the dataset. The x-axis shows the SHAP value, which reflects the impact of each feature on the model's prediction. Features with positive SHAP values increase the predicted outcome, whereas features with negative SHAP values lower the prediction. Each point in the plot corresponds to an individual data sample, with its color representing the feature value—blue for lower values and red for higher ones. Among all the features, Previous Scores exhibit the most significant impact on the model's output. Higher previous scores (represented in red) tend to increase the prediction, whereas lower scores (in

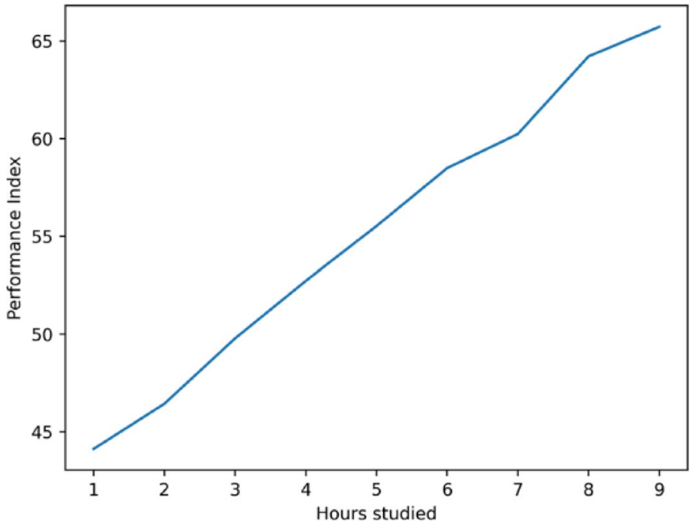

Fig. 6. Relationship between daily hours studied and average performance index. The plot illustrates a positive correlation, emphasizing the impact of study time on academic outcomes.

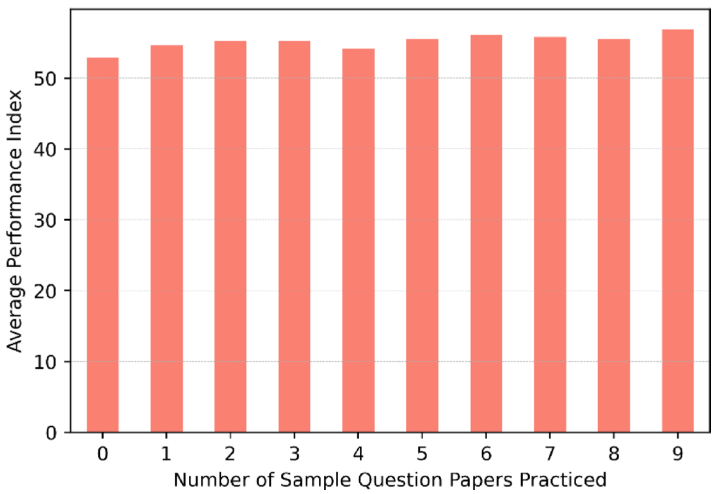

Fig. 7. Impact of the number of sample question papers practiced on average performance index. Students practicing around nine papers tend to achieve the highest performance.

Model	MAE	RMSE	R^2 Score
Linear Regression	0.0839	0.1052	0.9889
Ridge	0.0839	0.1052	0.9889
XGBRegressor	0.0862	0.1083	0.9883
CatBoosting Regressor	0.0864	0.1083	0.9883
AdaBoost Regressor	0.1183	0.1505	0.9774
Random Forest Regressor	0.0948	0.1186	0.9860
SVR	0.0861	0.1080	0.9884
K-Neighbors Regressor	0.1221	0.1519	0.9770
Bagging Regressor	0.1267	0.1573	0.9754
Ensemble VR	**0.0837**	**0.1050**	**0.9890**

Table 8. Results of different regression approaches using all features.

Model	MAE	RMSE	R^2 Score
XGBoost	0.0869	0.1095	0.988
CatBoost	0.0873	0.1099	0.9879
SVR	0.0864	0.1089	0.9881
Ensemble VR	**0.0841**	**0.1059**	**0.9888**

Table 9. Results of different approaches using all features and cross validation (first dataset)

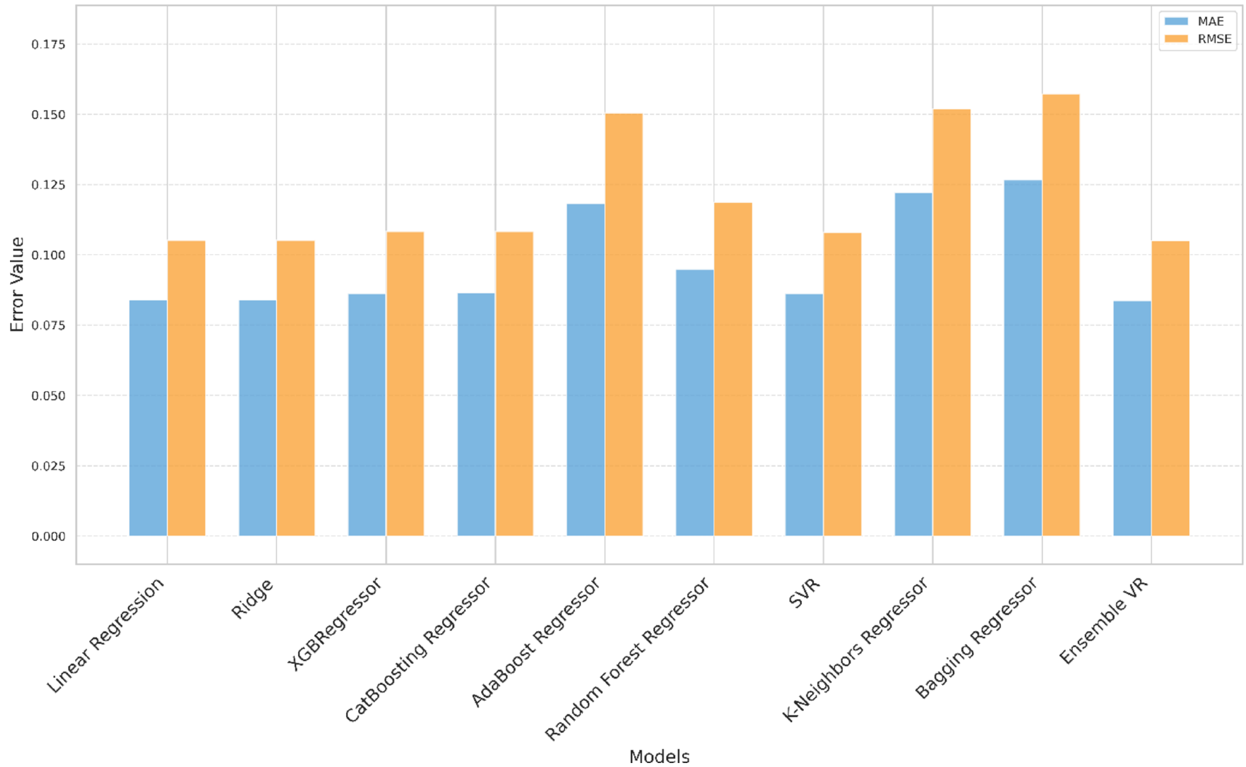

Fig. 8. Comparison of the model's MAE and RMSE errors using all features. Lower values indicate better model performance.

blue) contribute negatively. Hours Studied also plays a crucial role, with more study hours positively influencing the prediction, while fewer study hours decrease it. The impact of Sleep Hours appears to be more complex, as both very high and very low values affect predictions in different ways. This suggests that an optimal range of sleep might be beneficial, while too much or too little sleep could have mixed effects. Sample Question Papers Practiced generally contribute positively to the model's predictions, but their effect is not as strong as that of previous scores or study hours. Finally, Extracurricular Activities show the least influence on the prediction, with most SHAP values clustered around zero. Figure 16 demonstrates the interpretation of LIME's prediction for the first instance, utilizing five selected features to provide an explanation for the model's output.

Discussion
The efficacy metrics for various regression models when utilizing all available features are presented in Table 8. The results indicate that Ensemble VR stands out slightly as the best-performing model with the lowest MAE (0.0837), lowest RMSE (0.1050), and highest R2 Score (0.9890). The Linear Regression and Ridge Regression exhibit exceptional performance, as demonstrated by their high R^2 scores and low error metrics (MAE and RMSE). This suggests that these models effectively capture linear relationships within the data, resulting in a high degree of predictive accuracy. Compared to these models, XGBRegressor and CatBoosting Regressor also demonstrate strong predictive capabilities, albeit with slightly higher error metrics. The robustness of these models is reflected in their consistently high R^2 score of 0.9883. In contrast, the AdaBoost Regressor exhibits a considerable increase in error metrics, with elevated MAE and RMSE values suggesting that the model struggles to capture underlying data patterns effectively. The Random Forest Regressor, which operates as an ensemble method based on decision trees, delivers satisfactory performance but is slightly less precise than the best-performing models. The ensemble approach provides an advantage by reducing variance and enhancing robustness. Similarly, the SVR achieves performance metrics comparable to those of the XGBRegressor and

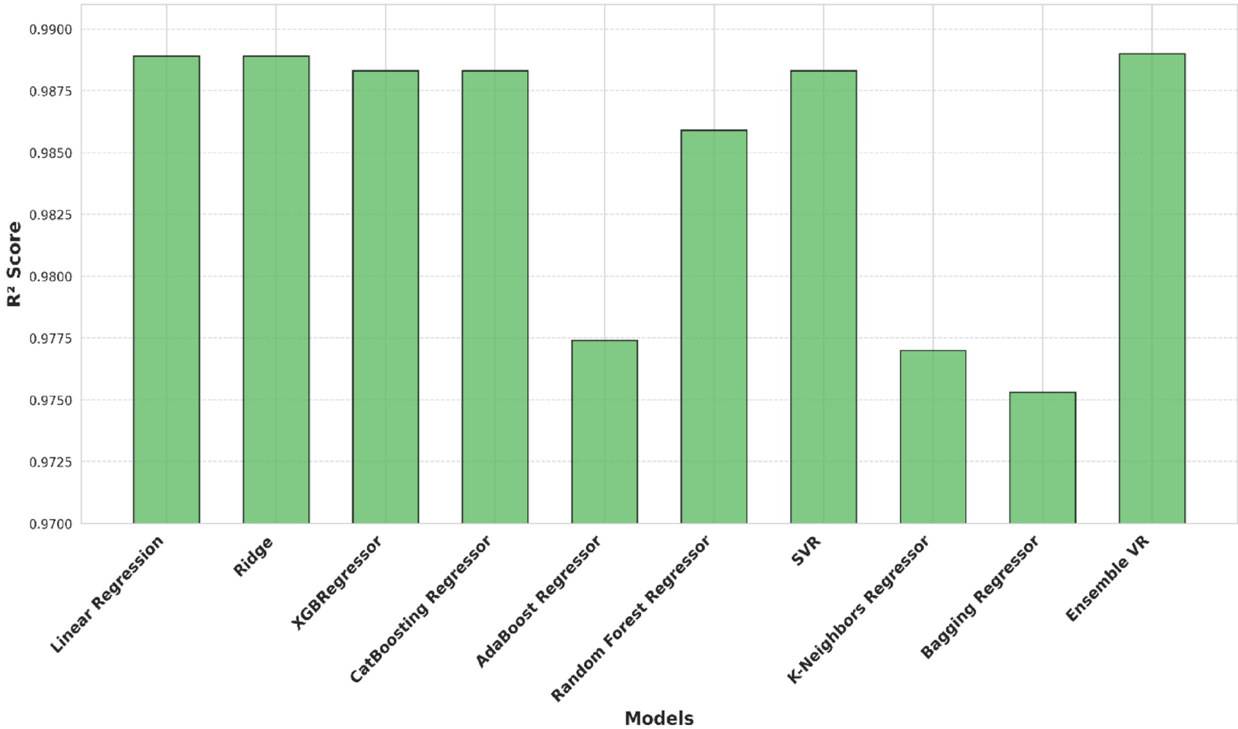

Fig. 9. Comparison of the model's R^2 scores using all features. Higher R^2 values indicate stronger explanatory power and better model fit.

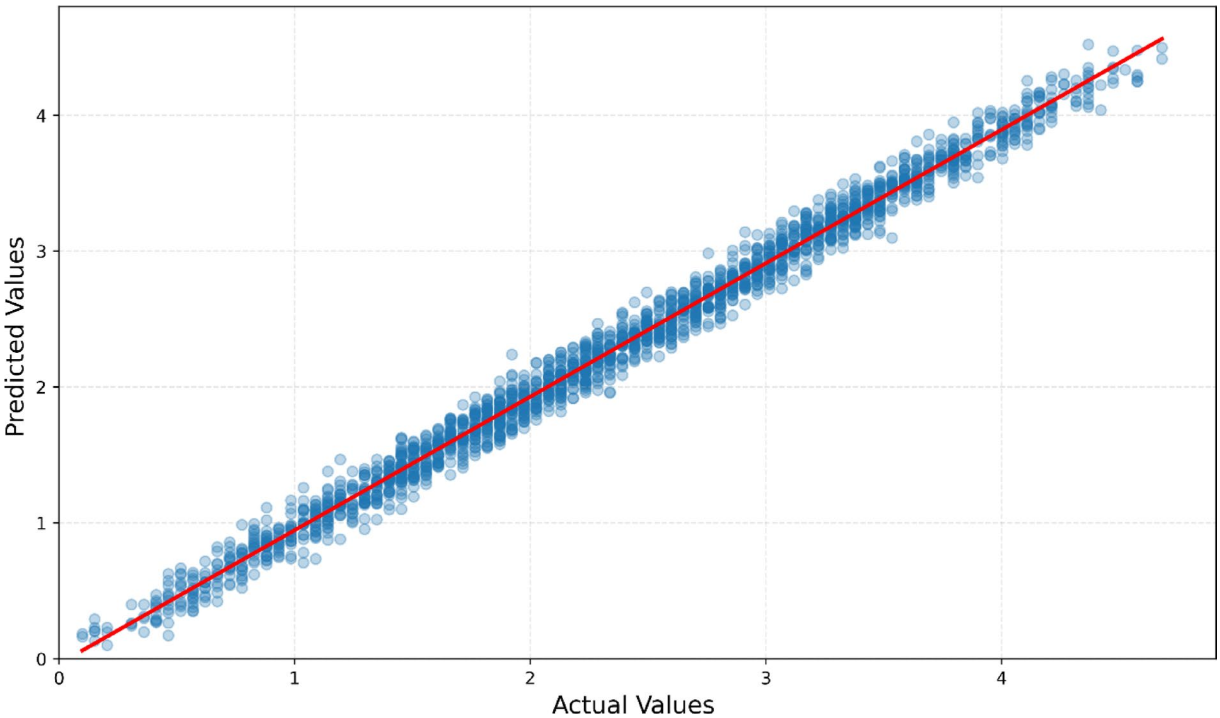

Fig. 10. Scatter plot of actual vs. predicted performance index values using the Ensemble Voting Regressor with all features. Points clustering along the diagonal line indicate high prediction accuracy.

Model	MAE	RMSE	R^2 Score
Linear Regression	0.0951	0.1192	0.9859
Ridge	0.0951	0.1192	0.9859
XGBRegressor	0.0956	0.1202	0.9856
CatBoosting	0.0962	0.1206	0.9855
AdaBoost	0.1183	0.1505	0.9774
Random Forest Regressor	0.0975	0.1224	0.9851
SVR	0.0951	0.1195	0.9858
K-Neighbors Regressor	0.1049	0.1319	0.9827
Bagging Regressor	0.1267	0.1573	0.9754
Ensemble VR	**0.0949**	**0.1190**	**0.9861**

Table 10. Results of different regression approaches using feature selection (first dataset).

Model	MAE	RMSE	R^2 Score
Linear Regression	0.4442	1.7994	0.7709
Ridge	0.4442	1.7994	0.7709
XGBRegressor	1.0102	2.2431	0.6440
CatBoosting	0.6403	1.9437	0.7327
Random Forest Regressor	1.0721	2.1575	0.6707
SVR	0.5709	1.8614	0.7549
K-Neighbors Regressor	1.6012	2.5964	0.5231
Bagging Regressor	1.2238	2.3211	0.6188
Ensemble VR	**0.4430**	**1.7981**	**0.7716**

Table 11. Results of different regression approaches using all features (second dataset).

Model	MAE	RMSE	R^2 Score
Linear Regression	1.2304	2.2132	0.6535
Ridge	1.2304	2.2132	0.6535
XGBRegressor	1.5184	2.6657	0.4973
CatBoosting	1.3424	2.3574	0.6068
Random Forest Regressor	1.3970	2.4295	0.5824
SVR	1.2417	2.2329	0.6473
K-Neighbors Regressor	1.4737	2.4722	0.5676
Bagging Regressor	1.5547	2.6638	0.4980
Ensemble VR	**1.2300**	**2.2125**	**0.6540**

Table 12. Results of different regression approaches using feature selection (second dataset).

CatBoosting Regressor, indicating its ability to detect fundamental data patterns effectively. However, the K-Neighbors Regressor substantially increases error metrics compared to other models, suggesting that it may not be well-suited for this dataset. A similar pattern is observed with the Bagging Regressor, which demonstrates a lower R^2 score and higher error metrics, reflecting its reduced efficiency. The best overall performance is observed with the proposed Ensemble VR, which attains the highest R^2 score of 0.9890 and the lowest error metrics (MAE of 0.0837, RMSE of 0.1050). This result highlights the effectiveness of combining multiple models to enhance predictive accuracy by leveraging their strengths. The ensemble approach proves beneficial in minimizing both bias and variance, leading to a more robust predictive model.

Table 9 presents the impact of cross-validation on the proposed model's performance. The Ensemble VR model with cross-validation (MAE: 0.0841, RMSE: 0.1059, R^2: 0.9888) shows a minimum difference in performance compared to the model without cross-validation. This difference indicates that the Ensemble VR model generalizes extremely well to unseen data, and the proposed model is robust and not overfitting.

We perform paired t-tests on the RMSE values from the 10-fold cross-validation using the results from Table 9. The p-value < 0.05 for all three models (SVR, XGBoost, and CatBoost) indicates that the larger RMSE gaps (0.003–0.004) indicate that the voting regressor reliably outperforms these models.

The impact of feature selection on model performance is presented in Table 10. The results reveal that employing feature selection generally leads to an increase in error metrics across all models.

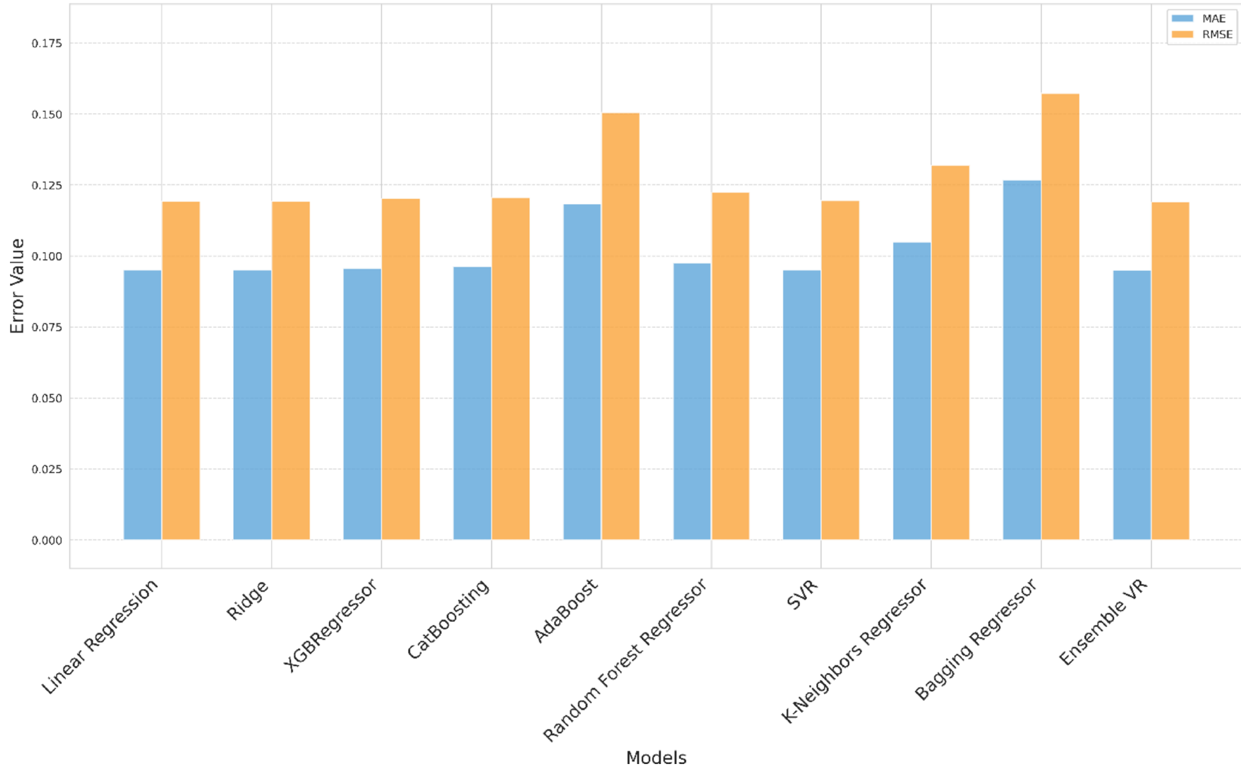

Fig. 11. Comparison of the model's MAE and RMSE errors using feature selection.

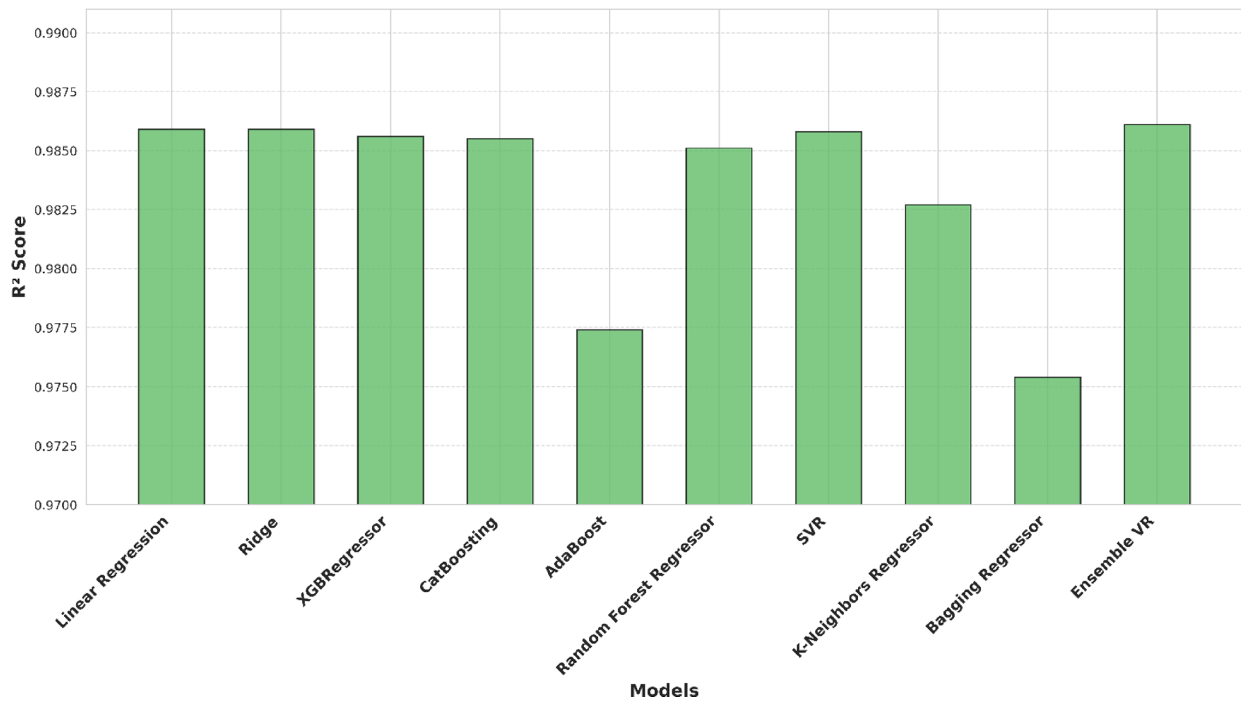

Fig. 12. Comparison of the model' R^2 scores using feature selection. A slight reduction in R^2 indicates that selected features explain less variance than the full feature set.

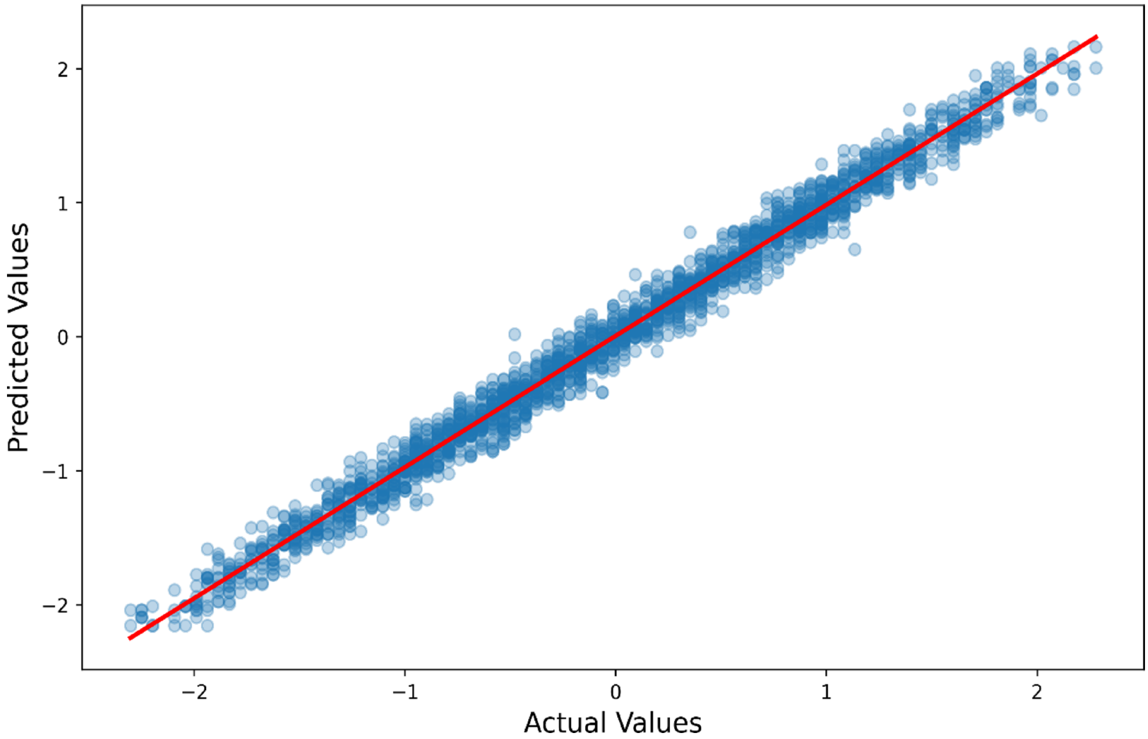

Fig. 13. Actual vs. predicted performance index using the proposed model after feature selection. Despite some degradation, the predictions remain close to the actual values.

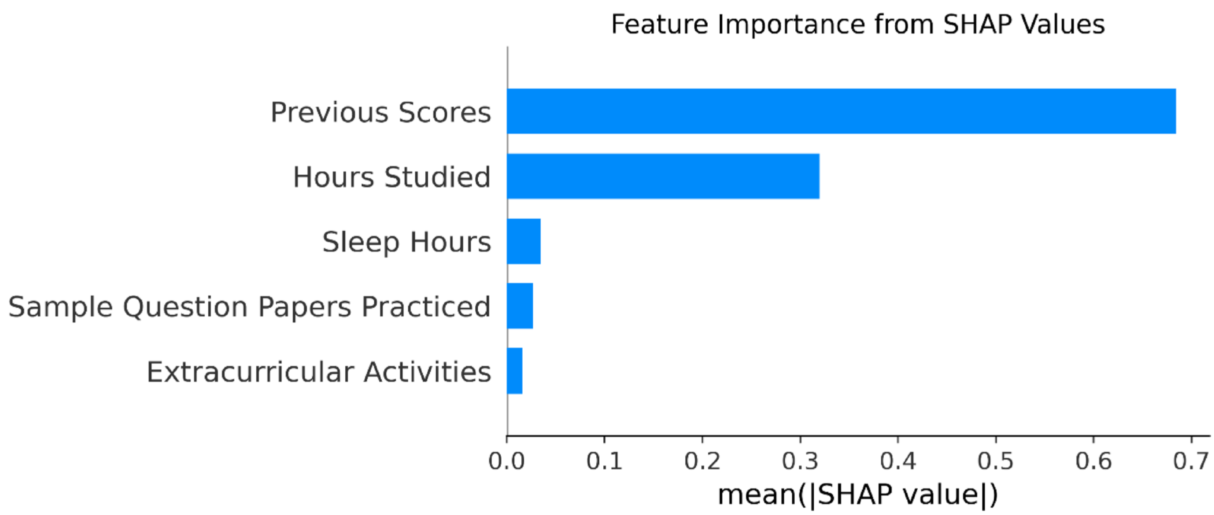

Fig. 14. Average SHAP values showing the global importance of each feature in the ensemble model. 'Previous Scores' is the most influential feature in predicting academic performance.

The Linear Regression and Ridge Regression models exhibit higher MAE (0.0951) and RMSE (0.1192) values than when all features are employed, and their R^2 scores also decrease (0.9859), suggesting a decrease in performance as a result of feature selection. The XGBRegressor and CatBoosting models exhibit a slight increase in MAE and RMSE and a decrease in R^2 scores. This implies that the model's performance is influenced by removing certain informative features due to feature selection. Random Forest exhibits a decrease in R^2 score (0.9851) and a slight increase in MAE (0.0975) and RMSE (0.1224). The selection of features results in a slight decrease in the performance of SVR. The K-Neighbors Regressor exhibits a slight decrease in MAE (0.1049) and RMSE (0.1319), but a lower R^2 score (0.9827). The Bagging Regressor's performance continues to be low. The Ensemble VR model proposed is the most resilient, as it maintains the lowest error metrics and the highest R^2 score, despite the challenges posed by feature selection. Nevertheless, its performance is inferior to that of the ensemble when all features are employed, suggesting that the ensemble's efficacy is influenced by feature

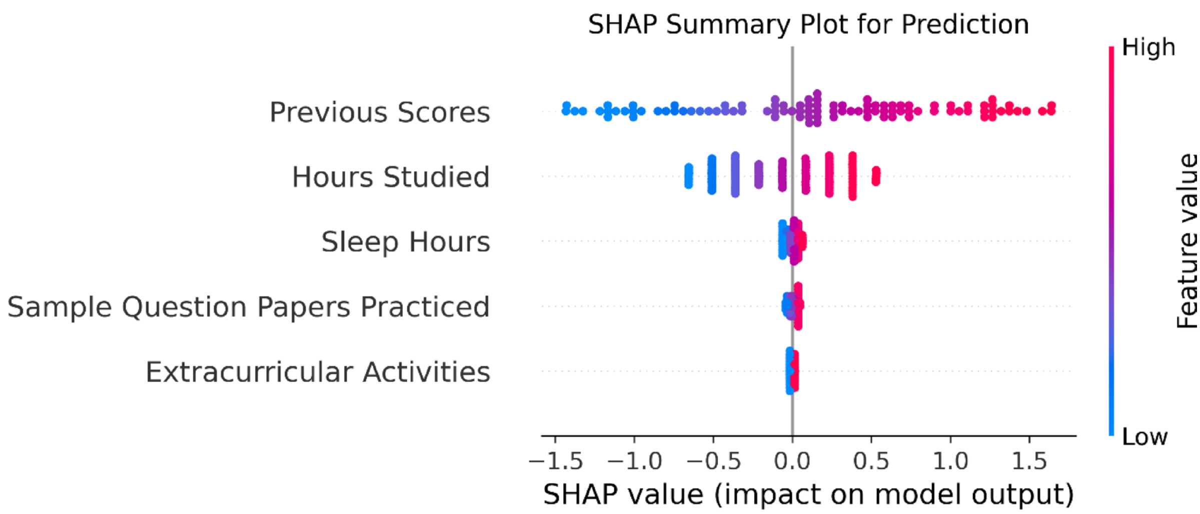

Fig. 15. SHAP summary plot shows how each feature impacts model predictions. Red indicates high feature values, blue indicates low. Features to the right increase predicted performance index; those to the left decrease it.

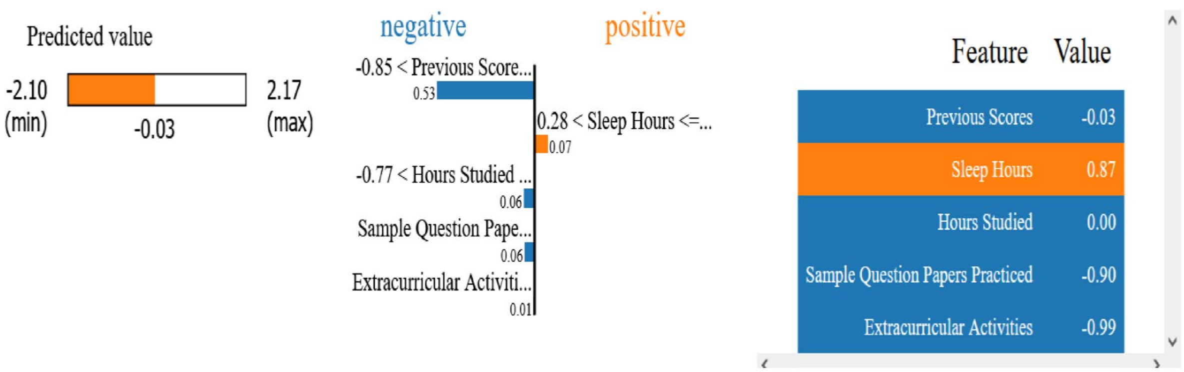

Fig. 16. LIME explanation for a sample prediction. The bar chart shows positive and negative contributions of individual features to the model's prediction for one student.

selection. From Table 11, the models work well when all the features are included. The Linear Regression, Ridge Regression, and VR models are the best. All three have R^2 ratings of about 0.77. The MAE values for these models were lower than other models. The SVR model also did well (R^2 = 0.7549, MAE = 0.5709), and CatBoosting did well too (R^2 = 0.7327). But models such as K-Neighbors Regressor, Bagging Regressor, Random ForestRegessor and XGBRegressor did not do as well as the best ones. From Table 12, When the feature selection is used, the model's performance decreases on all measures. The R^2 values for Linear and Ridge Regression reduced to 0.6535, and the MAE increased to 1.2304. The Ensemble VR continued to perform better than others with the highest R^2 (0.6540). SVR held consistent performance (R^2 = 0.6473). Tree model behavior, such as Random Forest, CatBoost, and XGBoost, was more affected by the reduced feature set. The ensemble VR achieves the optimal balance between robustness and predictive accuracy, resulting in an R^2 of 0.7716 when trained on the complete feature set. Reducing features was largely unhelpful in addressing this task, as evidenced by the precipitous decline in performance observed after feature selection, with R^2 declining to 0.6540.

Figure 14 further confirms the effectiveness of the proposed model under these conditions. Feature importance analysis using SHAP values is illustrated in Fig. 15. Features are ranked based on their influence on model predictions, with higher mean SHAP values indicating greater significance. Previous Scores emerge as the most influential feature, displaying the highest variation in SHAP values. The number of Hours Studied also exhibits a clear pattern, positively impacting predictions as study time increases. In contrast, Sleep Hours, Sample Question Papers Practiced, and Extracurricular Activities have more narrowly distributed SHAP values centered near zero, indicating minimal influence on predictions.

Further interpretability is provided through LIME, as illustrated in Fig. 16. LIME generates a local surrogate model to explain individual predictions, identifying both positive and negative feature impacts. For instance, Previous Scores, Hours Studied, and Sample Question Papers Practiced negatively impact some predictions, whereas Sleep Hours exhibits a positive effect. These insights highlight the nuanced impact of different features on predictive outcomes. The results demonstrate that models perform better when all features are included rather than when feature selection is applied. This underscores the importance of supplementary information provided by all features in enhancing predictive performance. The proposed Ensemble VR model stands out as the most effective, maintaining high accuracy and robustness under both conditions.

The significant advantages of this approach include:

- The inclusion of multiple regression algorithms provides diverse perspectives and highlights the trade-offs between simplicity, interpretability, and predictive accuracy.
- By employing a voting ensemble regression model to predict student performance, we can capitalize on the strengths of multiple models to improve the accuracy, robustness, and interpretability of the predictions.
- The analysis identifies key factors influencing student performance, such as extracurricular participation, sleep hours, and preparation levels, which can inform targeted interventions.
- The methodology can be adapted to larger datasets and different educational contexts, making it a versatile tool for stakeholders.
- The proposed voting Regression models offer clear insights into the impact of individual features, aiding educators and policymakers in decision-making.
- The detailed insights provided by LIME and SHAP enable educators to design targeted strategies for students who are struggling.
- The proposed methodology significantly enhances the accuracy, interpretability, and support of informed decision-making in predicting student performance in real-world educational environments.

Despite its promising results, this study has certain limitations. While the ensemble approach enhances predictive accuracy, further improvements could be achieved by incorporating more advanced stacking models to increase the R^2 score and reduce error metrics. Additionally, the interpretability of deep learning models remains a challenge, requiring further exploration of explainability techniques such as SHAP and LIME. Future research should focus on integrating more sophisticated deep learning architectures, such as transformer-based regression models, which may further enhance predictive performance. Additionally, exploring the impact of external factors such as socioeconomic background, teacher effectiveness, and institutional quality can provide a more holistic understanding of student performance prediction. Lastly, incorporating real-time adaptive learning recommendations based on model predictions can contribute to personalized education strategies, ultimately improving student outcomes. The proposed methodology provides a strong foundation for further advancements in educational analytics. By refining predictive models and expanding the scope of features considered, future studies can contribute to more effective and actionable insights for educators and policymakers. Although feature selection helped in reducing model complexity, it was observed that in some cases, it negatively affected model accuracy due to the potential removal of relevant features. For instance, the voting regression model yielded slightly better performance using all features compared to the reduced subset, suggesting that critical information might have been omitted during selection. This trade-off highlights the need for a more nuanced approach to feature selection. In future work, we plan to explore alternative dimensionality reduction techniques such as Principal Component Analysis (PCA), t-SNE, and autoencoders to better capture latent structure while minimizing information loss. Moreover, incorporating domain-driven feature engineering strategies could provide a more informative and compact representation of the data, potentially improving model generalizability.

While deep learning has been successful in educational analytics, our study utilizes data (10,000 rows × 6 features) that exhibits clear, interpretable relationships between variables. In such cases, our proposed model based on machine learning often remains competitive due to its simplicity, computational efficiency, and robustness to overfitting on smaller datasets. While the proposed methodology leverages classical machine learning regression models, recent studies have demonstrated the potential of deep learning architecture such as multilayer perceptron (MLPs), convolutional neural networks (CNNs), and recurrent neural networks (RNNs)—for predicting student performance, particularly when working with large-scale temporal or behavioral datasets. These models can automatically learn complex feature interactions without explicit feature engineering but often require significantly larger datasets, higher computational resources, and pose challenges in terms of interpretability. Given our dataset's size and the emphasis on model transparency in educational decision-making, classical regression models were more appropriate for this study. However, there is substantial potential for future research to benchmark our ensemble model against deep learning approaches, particularly as data complexity and volume increase.

Conclusion and future research directions

Understanding the factors influencing student performance in academic institutions is a critical area of research with significant implications for enhancing educational practices and outcomes. In this study, we proposed a robust methodology to predict student performance using a combination of preprocessing techniques and multiple regression models. Our approach evaluated the efficacy of nine algorithms, along with a voting algorithm based on the top five models, using performance metrics such as MAE, RMSE, and R^2 score. The results demonstrated that the proposed Ensemble Voting Regressor achieved the best performance, with an MAE of 0.0837, RMSE of 0.1050, and R^2 score of 0.9890 when using all features. The Ridge and Linear Regression models also performed competitively, each achieving an R^2 score of 0.9889. The incorporation of interpretable models such as LIME and SHAP allowed for better transparency, enabling us to identify key predictors of student success, such as participation in extracurricular activities, hours of sleep, and preparation levels. Future research will focus on several key directions:

- Exploring deep learning models and ensemble learning techniques to further improve prediction performance if the data is extended.
- Incorporating socioeconomic factors, parental involvement, and psychological attributes to develop a more holistic model of academic success.
- Developing adaptive learning systems that provide real-time feedback to students and educators based on predictive insights.
- Investigating potential biases in predictive models and ensuring fair, unbiased decision-making in educational analytics.

This study lays the groundwork for innovative approaches to educational data analytics, demonstrating the practicality of ensemble learning methods in predicting student performance. By leveraging interpretable AI techniques, this research provides actionable insights that can support data-driven improvements in teaching and learning processes. Although the dataset used in this study is publicly available and anonymized, we recognize the critical importance of protecting student privacy and ensuring ethical handling of educational data. Future applications of predictive models in real-world educational contexts must adhere to strict data governance policies, safeguard individual privacy, and ensure transparency and fairness in decision-making.

Data availability
Data are available at https://www.kaggle.com/datasets/mitgandhi10/dataset-for-multiple-regression. Accessed [08-04-2025].

Received: 8 April 2025; Accepted: 16 July 2025
Published online: 24 July 2025

References
1. Lau, E. T., Sun, L. & Yang, Q. Modelling, prediction and classification of student academic performance using artificial neural networks. *SN Appl. Sci.* **1** (9), 982 (2019).
2. Adewale, M. D., Azeta, A., Abayomi-Alli, A. & Sambo-Magaji, A. Empirical investigation of multilayered framework for predicting academic performance in open and distance learning. *Electronics* **13** (14), 2808 (2024).
3. Li, X. & He, S. Research and analysis of student portrait based on campus big data. In 2021 IEEE 6th International Conference on Big Data Analytics (ICBDA) (pp. 23–27). IEEE. (2021), March.
4. Yang, F. & Li, F. W. Study on student performance estimation, student progress analysis, and student potential prediction based on data mining. *Comput. Educ.* **123**, 97–108 (2018).
5. Kausar, S. et al. Mining smart learning analytics data using ensemble classifiers. *Int. J. Emerg. Technol. Learn. (iJET)*. **15** (12), 81–102 (2020).
6. Sarker, I. H. Machine learning: algorithms, real-world applications and research directions. *SN Comput. Sci.* **2** (3), 160 (2021).
7. Haron, N. H., Mahmood, R., Amin, N. M., Ahmad, A. & Jantan, S. R. An artificial intelligence approach to monitor and predict student academic performance. *J. Adv. Res. Appl. Sci. Eng. Technol.* **44** (1), 105–119 (2025).
8. Vimarsha, K., Prakash, S. S., Krinkin, K. & Shichkina, Y. A. Student performance prediction: A Co-Evolutionary hybrid intelligence model. *Procedia Comput. Sci.* **235**, 436–446 (2024).
9. Ahmed, E. Student performance prediction using machine learning algorithms. *Appl. Comput. Intell. Soft Comput.* **2024** (1), 4067721 (2024).
10. Chen, Z., Cen, G., Wei, Y. & Li, Z. Student performance prediction approach based on educational data mining. *IEEE Access.* **11**, 131260–131272 (2023).
11. Liu, D. et al. Multiple features fusion attention mechanism enhanced deep knowledge tracing for student performance prediction. *IEEE Access.* **8**, 194894–194903 (2020).
12. Mastour, H., Dehghani, T., Moradi, E. & Eslami, S. Early prediction of medical students' performance in high-stakes examinations using machine learning approaches. Heliyon, 9(7). (2023).
13. Xu, H. & Kim, M. Combination prediction method of students' performance based on ant colony algorithm. *Plos One*, **19**(3), e0300010. (2024). https://doi.org/10.1371/journal.pone.0300010
14. Gandhi, M. Dataset for Multiple Regression: A Comprehensive Dataset for Predicting and Analyzing Student Performance, Kaggle. [Online]. Available: https://www.kaggle.com/datasets/mitgandhi10/dataset-for-multiple-regression. [Accessed: Jan. 14, 2025].
15. Lai Ng Dataset for Multiple Regression: A Comprehensive Dataset for Predicting and Analyzing Student Performance, Kaggle. [Online]. Available: https://www.kaggle.com/datasets/lainguyn123/student-performance-factors/data
16. Alshanqiti, A. & Namoun, A. Predicting student performance and its influential factors using hybrid regression and multi-label classification. *Ieee Access.* **8**, 203827–203844 (2020).
17. Cunningham, P. & Delany, S. J. K-nearest neighbour classifiers-a tutorial. *ACM Comput. Surv. (CSUR)*. **54** (6), 1–25 (2021).
18. Sabry, F. *K Nearest Neighbor Algorithm: Fundamentals and Applications*Vol. 28 (One Billion Knowledgeable, 2023).
19. Cheng, B., Liu, Y. & Jia, Y. Evaluation of students' performance during the academic period using the XG-Boost Classifier-Enhanced AEO hybrid model. *Expert Syst. Appl.* **238**, 122136 (2024).
20. Asselman, A., Khaldi, M. & Aammou, S. Enhancing the prediction of student performance based on the machine learning XGBoost algorithm. *Interact. Learn. Environ.* **31** (6), 3360–3379 (2023).
21. Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A. V. & Gulin, A. CatBoost: unbiased boosting with categorical features. *Adv. Neural. Inf. Process. Syst.*, **31**, (2018).
22. Schapire, R. E. Explaining adaboost. In *Empirical Inference: Festschrift in Honor of Vladimir N. Vapnik* 37–52 (Springer, 2013).
23. Banfield, R. E., Hall, L. O., Bowyer, K. W. & Kegelmeyer, W. P. A comparison of decision tree ensemble creation techniques. *IEEE Trans. Pattern Anal. Mach. Intell.* **29** (1), 173–180 (2006).
24. Zhang, F. & O'Donnell, L. J. Support vector regression. In Machine Learning (123–140). Academic. (2020).
25. Demir-Kavuk, O., Kamada, M., Akutsu, T. & Knapp, E. W. Prediction using step-wise L1, L2 regularization and feature selection for small data sets with large number of features. *BMC Bioinform.* **12**, 1–10 (2011).
26. Todorovski, L. & Džeroski, S. Combining multiple models with meta decision trees. In European conference on principles of data mining and knowledge discovery (pp. 54–64). Berlin, Heidelberg: Springer Berlin Heidelberg. (2000), September.
27. Meshoul, S., Batouche, A., Shaiba, H. & AlBinali, S. Explainable multi-class classification based on integrative feature selection for breast cancer subtyping. *Mathematics* **10** (22), 4271 (2022).
28. Karunasingha, D. S. K. Root mean square error or mean absolute error? Use their ratio as well. *Inf. Sci.* **585**, 609–629 (2022).

29. Chicco, D., Warrens, M. J. & Jurman, G. The coefficient of determination R-squared is more informative than SMAPE, MAE, MAPE, MSE and RMSE in regression analysis evaluation. *Peerj Comput. Sci.*, **7**, e623 (2021). https://doi.org/10.7717/peerj-cs.623/supp-1

Acknowledgements
This work is supported by Princess Nourah bint Abdulrahman University Researchers Supporting Project number (PNURSP2025R196), Princess Nourah bint Abdulrahman University, Riyadh, Saudi Arabia. The authors would like to acknowledge Prince Sultan University for their valuable support.

Author contributions
Wesam Ahmed: Methodology, Investigation, Review & Editing, Visualization. Mudasir Ahmad Wani: Conceptualization, Methodology, Investigation. Paweł Pławiak: Funding, Resources, Project administration. Souham Meshoul: Investigation, Validation, Funding, Review & Editing. Amena Mahmoud: Investigation, Validation. Mohamed Hammad: Methodology, Investigation, Review & Editing, Supervision. All authors read and approved of the final manuscript.

Funding
This work is supported by Princess Nourah bint Abdulrahman University Researchers Supporting Project number (PNURSP2025R196), Princess Nourah bint Abdulrahman University, Riyadh, Saudi Arabia.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

Additional information
Correspondence and requests for materials should be addressed to S.M. or M.H.